

WEEK 7: ORGAN SYSTEM

CARDIOVASCULAR SYSTEM

- A closed system consisting of heart, blood vessels, and blood

- Ensures continuous circulation of blood, which carries O₂, nutrients, hormones, and immune factors to tissues, and removes CO₂ and metabolic wastes

- Functions:

- Distribute essential substances - delivers O₂, nutrients, and hormones to tissues
- Removes metabolic wastes - carries CO₂, urea, and other byproducts to the lungs, kidneys, and liver for elimination
- Maintains BP and tissue perfusion - ensures adequate supply of O₂ and nutrients to meet metabolic demands
- Provide immune defense - transports WBC and antibodies
- Regulates body temperature and pH balance - via vasodilation/vasoconstriction and buffer transport

- In PT, blood flow and heart functions directly influence exercise tolerance, endurance, tissue healing, and overall patient mobility

BLOOD

- **Transport medium**

- Specialized connective tissue with a liquid extracellular matrix (plasma) and cellular elements (formed elements)

- Accounts for about **7-8% of total body weight (4-6L)**

- Functions:

- Transport of gases, nutrients, hormones, and wastes
- Regulation of body temperatures, pH, and fluid balance
- Protection against blood loss (**hemostasis = clotting**) and infection (**immune cells**)

- Composition:

- *Plasma (55%)*
 - **90%** - water

- **8%** - protein (**fibrinogen** - blood clotting, **albumin** - most abundant in the body, maintaining osmotic pressure and preventing fluid from leaking out of the blood vessels, **globulin** - transport of lipids and fat-soluble vitamins, and also includes antibodies to provide immune defense)

- **2%** - electrolytes, gases, nutrients, hormones, wastes

● *Formed elements (45%)*

- **Erythrocytes (RBCs)** - O₂ and CO₂ transport

- Biconcave discs, anucleate, 120-day lifespan
- **Packed with hemoglobin** - where oxygen binds
- Formed in bone marrow through **erythropoiesis** regulated by **erythropoietin** in kidney

- **Leukocytes (WBCs)** - defense

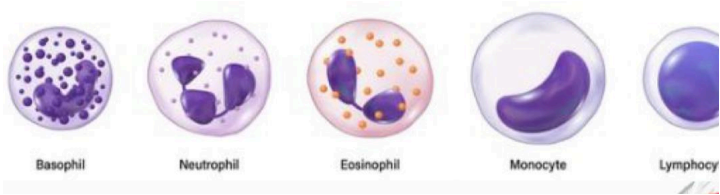
■ **Granulocytes**

- Neutrophils (phagocytosis, acute infection)
- Eosinophils (allergy, parasites)
- Basophils (histamine release) - produce inflammatory response

■ **Agranulocytes**

- Lymphocytes (B, T, NK cells)
- Monocytes (differentiate into macrophages) - **largest of all**

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platelets (through fragments of megakaryocytes). and granulocytes (through myeloblast)

- **Lymphoid stem cell** forms B, T, and NK cell through lymphoblast

○ Platelets (thrombocytes) - clotting

- Small, anucleated, disc-shaped cytoplasmic fragments derived from megakaryocytes in the bone marrow
- Each megakaryocyte can release thousands of platelets into the circulation
- **Normal platelet count:** 150,000 - 400,000/uL of blood
- **Average size:** 2-4 um in diameter
- **Lifespan:** 7-10 days; removed by the spleen and liver
- Lack nucleus, but contain mitochondria, microtubules, and actin filaments, and granules (MMAG) for movement and secretion
- Contain two main granules types
 - **Alpha:** platelet-derived growth factor (PDGF) - for tissue repair, clotting factors, fibrinogen
 - **Dense:** ADP, ATP, Ca, serotonin

- Hematopoiesis - blood cell formation

- **Starts from blood stem cell (hematopoietic stem cell) that is found in the red bone marrow in long bones**
- Branches into 2:
 - **Myeloid stem cell** to form red blood cells (through erythropoiesis),

HEART

- Hollow muscular organ that **acts as a double pump**, maintaining circulation through:

- **Pulmonary circulation** - sends deoxygenated blood to the lungs for gas exchange
- **Systemic circulation** - supplies oxygenated blood to the body

- Characteristics:

- **Size:** comparable to the owner's clenched fist (~300g)
- **Shape:** conical, with an **apex (inferolateral)** and a **base (posterior)**
- **Location:** middle mediastinum, obliquely placed between the lungs, $\frac{2}{3}$ left of the midline
- **Apex:** directed downward, forward, and to the left -> located at 5th intercostal space, midclavicular line
- **Base:** formed mainly by the left atrium (with contributions from the **right atrium**), points posteriorly directly from the spine -> 3rd costal cartilage (ribcage)

PERICARDIUM

- Fibrous pericardium

- Strong external layer; prevents overstretching and anchors heart to *diaphragm and sternum*

- Serous pericardium

- Has **parietal lining (lining fibrous sac)** and **visceral layer (epicardium)** adherent to the heart

- Pericardial cavity - between two pericardium

- Contains a thin film of serous fluid, reducing friction

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- **Pericardial effusion - too much pericardial fluid** -> contracts the heart and prevent it from filling properly
 - Causes cardiac tamponade
 - **Pericardio synthesis** - needle is inserted into the pericardial sac, usually at the left infrasternal angle to aspirate the excess fluid and relieve the pressure on heart

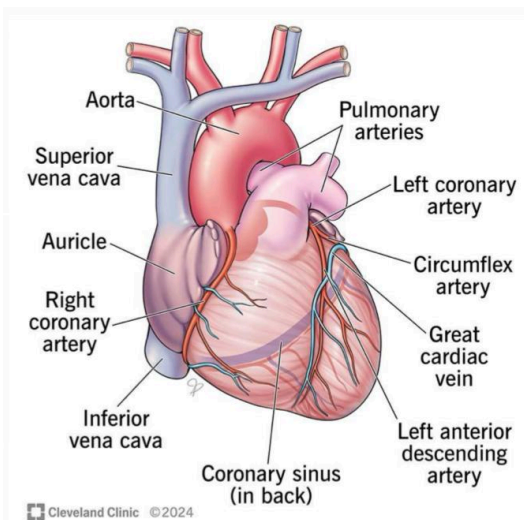
HEART CHAMBERS & VALVES

- **Right atrium:** receives systemic venous return (SVC, IVS, coronary sinus)
- **Right ventricle:** pumps deoxygenated blood into pulmonary trunk to the lungs for oxygenation
- **Left atrium:** receives oxygenated blood from the lungs through the pulmonary veins
- **Left ventricle:** **thickest myocardium**, pumps into systemic aorta to the body

Valves - prevents backflow

- **AV valves** - tricuspid valve (RA -> RV), mitral (LA -> LV)

- **Semilunar valves** - pulmonary (RV -> PA), aortic (LV -> aorta) - **ALWAYS STARTS WITH VENTRICLE**



EXTERNAL ANATOMY & CORONARY CIRCULATION

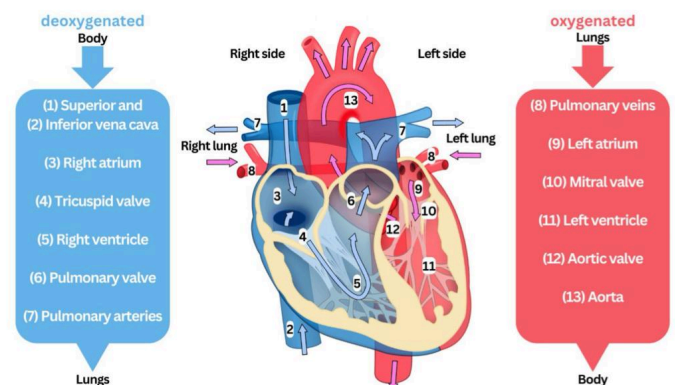
- Grooves (sulci):

- **Coronary sulcus** - separates atria from ventricles (up & down separation)
- **Anterior & posterior interventricular sulci** - mark ventricular septum (left & right separation)

- Coronary arteries (first branches of ascending aorta):

- **Left coronary artery (LCA)** - Only left
 - **Anterior interventricular (LAD)** - supplies anterior LV and interventricular septum
 - **Circumflex** - supplies LA & LV lateral wall
- **Right coronary artery (RCA)** - Left and posterior regions
 - **Marginal** - supplies RV
 - **Posterior interventricular** - supplies posterior septum & ventricles; arises from circumflex artery
- **Venous drainage:** coronary sinus - drains into right atrium

Path of Blood Flow Through the Heart



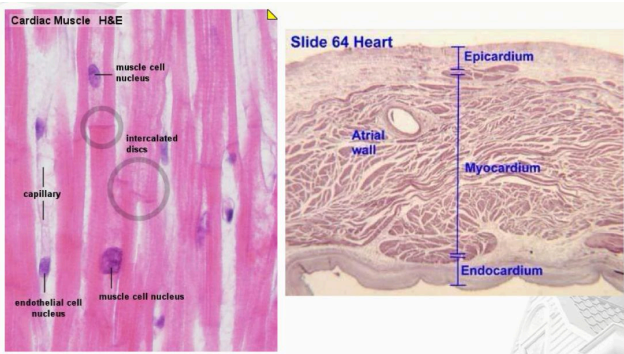
- Deoxygenated blood also originates at the coronary sinus

- **Pulmonary vein** - only vein the body that carries oxygenated blood

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HISTOLOGY

- **Epicardium** - visceral pericardium
- **Myocardium** - thick striated cardiac muscle layer; thicker than LV
- **Endocardium** - endothelium lining chambers & valves for blood flow without friction
- **Heart skeleton**
 - Dense fibrous framework surrounding valves
 - Functions:
 - Anchors valves
 - Provides rigid base for muscle contraction (prevents overstretching)
 - Electrically insulates atria from ventricles (critical for conduction) through AV node
- **Cardiac muscle**
 - Striated, involuntary, branched
 - **Intercalated discs**: contain gap junctions (electrical coupling) + desmosomes (mechanical strength)
 - Acts as a functional syncytium (contracts as one unit)



CONDUCTING SYSTEM

- **Sinoatrial node** (pacemaker, ~70-80 bpm)
 - Also called the natural pacemaker of the heart
 - **Location**: upper part of the RA, near the opening of the superior vena cava
 - **Structure**: a small specialized cluster of cells with inherent rhythmicity

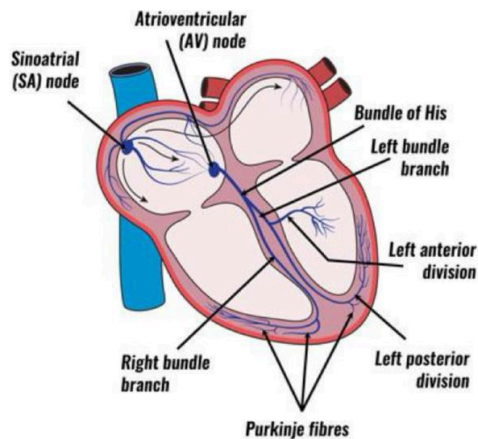
- **Function**:
 - Initiates the heartbeat by **spontaneous depolarization (automaticity)**
 - Sets the normal heart rate at 70-80 bpm in adults at rest
- **Clinical notes**: damage result in **arrhythmias** or slower heart rates, requiring artificial pacemaker
- **Atrioventricular node**
 - Delays conduction -> allows atrial emptying
 - From atria via SA node, it reaches AV node
 - **Location**: lower part of the interatrial septum, near the opening of the coronary sinus
 - **Function**:
 - Receives the impulses from the atria and delays the conduction (~0.1 sec)
 - This delay allows complete atrial contraction and emptying of blood into the ventricles before ventricular contraction begins
 - **Clinical note**: damage result in blockage of ventricle conduction (heart block), **leading to bradycardia** or dissociation of atrial and ventricular rhythms
- **Atrioventricular bundle (bundle of His)**
 - Bundle of His -> bundle branches -> Purkinje fibers
 - **Originates from the AV node**
 - Only normal electrical connection between the atria and ventricles (since the fibrous skeleton of the heart electrically insulates the atria from ventricles)
 - Travels through the interventricular septum, which descends into right and left bundle branches
 - Right and left bundle branches
 - Both ascended along the interventricular septum
 - **Right bundle**: continues toward the right ventricle, often passing through the moderator band

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- **Left bundle:** fans out more extensively across the left ventricle wall
- **Function:** conduct impulses rapidly forward toward the apex of the heart

- Purkinje fibers

- Fine, specialized conducting fibers branching **from the bundle branches into the ventricular myocardium**
- Large, fast-conducting cells that distribute the impulse to the ventricular muscle fibers
- Distribute the signal into the ventricular wall to start muscle contraction at the apex and then moves upward
- **Result:** coordinated contraction of ventricles, starting at the apex and progressing upward -> efficient ejection of blood into the pulmonary artery and aorta



HEART SOUNDS

- **S1 ("lub")** - AV valve closure
- **S2 ("dub")** - semilunar valve closure
- **S3, S4** - abnormal sound suggest pathology (e.g., heart failure)
- **Murmurs** - due to turbulent blood flow across defective valves

NORMAL HEART SOUNDS

1.) First heart sound (S1 - "lub")

- **Produced by:** closure of the AV valves -> mitral and tricuspid valves
- **Timing:** marks the onset of ventricular systole (isometric contraction)
- **Best heard at:** apex of the heart (5th intercostal space, midclavicular line)

2.) Second heart sounds (S2 - "dub")

- **Produced by:** closure of the semilunar valves -> aortic and pulmonary valves
- **Timing:** marks the end of systole and onset diastole
- **Best heard at:** base of the heart (3rd right costal spaces - for aortic, left for pulmonary)

EXTRA HEART SOUNDS (NORMALLY ABSENT IN ADULTS)

1.) Third heart sound (S3)

- Low-pitched sound **occurring just after S2 (early diastole)**
- **Caused by:** rapid ventricular filling against a compliant ventricle
- **Normal in:** children, young adults, and athletes
- **Abnormal in old adult:** heart failure, dilated cardiomyopathy, volume overload -> sometime called a "ventricular gallop"

2.) Fourth heart sound (S4)

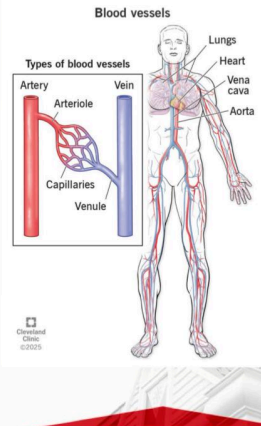
- Low-pitched sound **occurring just before S1 (late diastole)**
- **Caused by:** atrial contraction against a stiff or hypertrophied ventricle
- **Always abnormal in adults** -> associated with hypertension, aortic stenosis, hypertrophic cardiomyopathy
- Sometimes called an "atrial gallop"

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CIRCULATORY SYSTEM

Blood Vessel Structure

- **Arteries**
 - Thick, elastic, and muscular walls.
 - Designed to withstand **high pressure** generated by ventricular systole.
 - Carry blood **away from the heart** (usually oxygenated, except pulmonary arteries).
- **Capillaries**
 - Microscopic vessels composed of a **single endothelial layer**.
 - Function as the **exchange vessels** of gases, nutrients, and wastes between blood and tissues.
- **Veins**
 - Thinner walls compared to arteries; larger lumens.
 - Equipped with **valves** to prevent backflow.
 - Function as **capacitance vessels** or blood reservoirs, containing the majority of circulating blood.



Capillaries

- **Continuous Capillaries**
 - Found in muscle, brain, skin.
 - Least permeable, with tight junctions.
- **Fenestrated Capillaries**
 - Found in kidneys, endocrine glands, intestines.
 - Have pores (fenestrations) → allow rapid exchange of substances.
- **Sinusoids**
 - Found in liver, spleen, bone marrow.
 - Wide, irregular channels with gaps → permit passage of large molecules and even cells.

CAPILLARY TYPES

Lumen	Nucleus of endothelial cell	
Intercellular cleft	Basement membrane	Fenestrae
CONTINUOUS CAPILLARY	FENESTRATED CAPILLARY	DISCONTINUOUS CAPILLARY
Examples: nervous system and muscle.	Examples: glomerular and gut mucosa.	Examples: liver and marrow.

Types of Arteries

1. **Elastic Arteries** (e.g., aorta, pulmonary trunk)
 - Large arteries with high elasticity.
 - Function as **pressure reservoirs**: they expand during systole and recoil during diastole, maintaining continuous blood flow.
2. **Muscular Arteries**
 - Medium-sized arteries (e.g., femoral, brachial).
 - Contain more smooth muscle → distribute blood to specific organs.
3. **Arterioles**
 - Small arteries that regulate blood flow into capillary beds.
 - Known as **resistance vessels**, since changes in their diameter strongly influence blood pressure.

Major Arteries

1. **Aorta**
 - Largest artery, with four parts: ascending, arch, thoracic, abdominal.
2. **Carotid System**
 - **Internal carotid arteries** → supply brain and eyes.
 - **External carotid arteries** → supply face, scalp, and superficial structures.
3. **Vertebral Arteries**
 - Ascend through transverse foramina of cervical vertebrae.
 - Unite to form the **basilar artery**, which supplies the posterior brain.
4. **Coronary Arteries**
 - Arise from the ascending aorta.
 - Supply the myocardium itself with oxygenated blood.

Veins

- Begin as **venules** (small vessels draining capillaries).
- Merge to form larger veins → eventually lead into the **superior and inferior vena cava**.
- Features:
 - **Valves** prevent backflow, especially in limbs.
 - Act as **blood reservoirs**, holding about **65% of blood volume at rest**.

“VACC”

1.) Aorta - AAAT

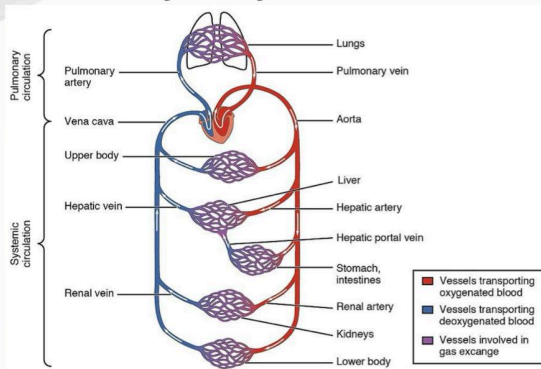
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Vein Characteristics

- **Low-pressure system** due to distance from the pumping action of the heart.
- **Thin-walled** with larger lumens compared to arteries.
- Venous return aided by:
 - **Skeletal muscle pump** (muscle contraction squeezes veins).
 - **Respiratory pump** (pressure changes during breathing help move blood).
- **Valves** prevent backflow, particularly in the limbs.

Veins - collect used blood (deoxygenated)

Pulmonary vs. Systemic Circulation



Pulmonary (low pressure system) - blood leaving the RV travels through pulmonary arteries through the lungs where it becomes oxygenated. It returns to LA via pulmonary veins

Systemic (high pressure system) - starts with LV, which pumps oxygenated blood into the aorta. From there, blood travels to arteries, arterioles, and capillaries to deliver it to tissues

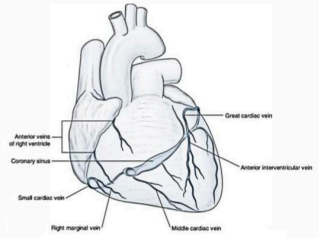
VENOUS DRAINAGE OF THE HEART

- The **myocardium** requires constant oxygen supply and removal of metabolic wastes.
- While the **coronary arteries** supply oxygenated blood, the **cardiac veins** provide the essential return pathway, draining deoxygenated blood back to the **right atrium**.
- Venous drainage occurs mainly through the **coronary sinus system**, with accessory drainage through the **anterior cardiac veins** and **smallest cardiac veins (Thebesian veins)**.

Clinical importance: Venous occlusion, like arterial blockage, can impair myocardial function. Knowledge of venous pathways is crucial for **cardiac imaging, bypass surgery, and electrophysiological procedures**.

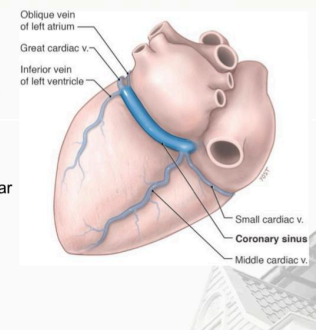
Main Venous Channels

1. **Coronary sinus and its tributaries** (major drainage system, ~60–70% of venous return).
2. **Anterior cardiac veins** (direct to right atrium).
3. **Smallest cardiac veins (Thebesian veins)** (direct into chambers).



Coronary Sinus

- **Location:** Posterior coronary sulcus, between the left atrium and left ventricle.
- **Termination:** Opens into the right atrium between the opening of the inferior vena cava (IVC) and the tricuspid orifice.
- **Valve:** The **Thebesian valve** (semicircular fold of endocardium) guards its opening.
- **Function:** Acts as the main collecting vessel for venous blood from the myocardium.

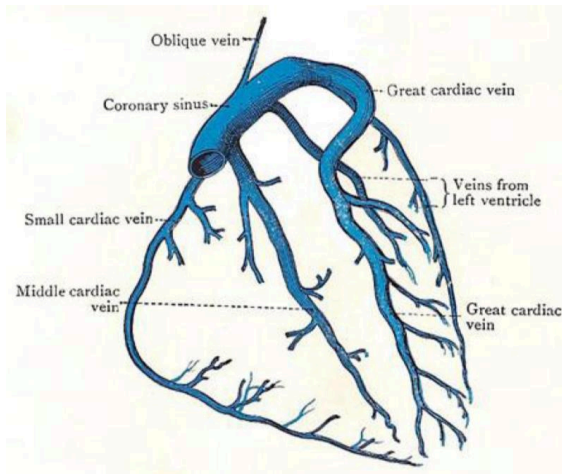


Tributaries of the Coronary Sinus

1. **Great Cardiac Vein**
 - Largest tributary.
 - Runs in the anterior interventricular sulcus with the anterior interventricular artery (LAD).
 - Curves to the left in the coronary sulcus, ending in the coronary sinus.
 - Drains the left atrium, both ventricles.
2. **Middle Cardiac Vein**
 - Runs in the posterior interventricular sulcus with the posterior interventricular artery.
 - Drains most of the posterior part of the heart.
3. **Small Cardiac Vein**
 - Runs in the right atrioventricular (coronary) sulcus with the right coronary artery.
 - Drains the right atrium and right ventricle.
4. **Posterior Vein of the Left Ventricle**
 - Ascends on the diaphragmatic surface of the left ventricle.
 - Drains into the coronary sinus.
5. **Oblique Vein of the Left Atrium (Vein of Marshall)**
 - A small vein descending obliquely on the back of the left atrium.
 - Remnant of the left common cardinal vein of the embryo.
 - Joins the coronary sinus near its left extremity.

Small cardiac vein - right coronary artery (right marginal artery or post. interventricular)

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Anterior Cardiac Veins

- Several small veins on the **anterior surface of the right ventricle**.
- Unlike other veins, they **do not drain into the coronary sinus**, but open **directly into the right atrium**.
- Functionally important in providing **additional drainage** for the right side of the heart.

Smallest Cardiac Veins (Thebesian Veins)

- Minute veins located within the walls of all four chambers.
- Open **directly into the chambers** of the heart, particularly the **right atrium and right ventricle**.
- More numerous on the right side of the heart.

Physiological significance:

- They contribute to the **physiological right-to-left shunt**, because a small volume of deoxygenated blood from the Thebesian veins drains directly into the left atrium and left ventricle, mixing with oxygenated blood.
- Clinically relevant in arterial blood gas measurements, as it explains why systemic arterial PO_2 is slightly less than alveolar PO_2 even in healthy individuals.

Clinical Anatomy Correlations

1. **Cardiac Catheterization**
 - The **coronary sinus** is used as an access point in procedures like biventricular pacemaker implantation or electrophysiological mapping.
2. **Coronary Venous Thrombosis**
 - Though rare, thrombosis of the coronary sinus or its tributaries can impair myocardial venous return, contributing to ischemia.
3. **Coronary Sinus Dilatation**
 - Seen in conditions with increased right atrial pressure (e.g., tricuspid valve disease, right heart failure).
4. **Vein of Marshall**
 - An important landmark in cardiac ablation procedures for atrial fibrillation.
5. **Thebesian Veins**
 - Their direct drainage into chambers contributes to the **normal A-a gradient** in blood gas analysis.

- Schemia - shortage of O_2 in heart tissue (irregular heartbeat)